a
Science
Kidney multiome-based genetic scorecard reveals convergent coding and regulatory variants
Liu H, ... Susztak K.
Science 2025
doi:10.1126/science.adp4753
Title, DOI, PMID or citation
igvf-replicate-paper
IGVF Agent
liu2025-open4gene: hurdle negative-binomial peak→gene test (port of R/Open4Gene.R; zero-model β and P identical)
encode-re2g: ENCODE-rE2G enhancer→gene model, 9 pretrained (max |Δscore| 7.8e-16 vs the authors' K562 test)
esig_net_predict: eSIG-Net inference on the released checkpoint (bit-identical to the authors' test.sh)
Deposited data: Figshare, Supp. Tables, Source Data
Authors' code at pinned commits, run as reference
Connect to an agent
IGVF Agent
User: Replicate Liu et al. 2025, Science (doi:10.1126/science.adp4753)
Agent: 11 of 16 analyses reproduced.
Fig. 3D re-drawn: R = 0.520 (paper 0.51).
2 blocked: individual-level data (controlled access).
Ported methods (CLI tools)
Data
Reference code
b
/igvf-replicate-paper Liu et al. 2025 Science
1 igvfagent bench resolve → PMID 39913582 · PMC12013656
2 igvfagent bench harvest → Figshare 26299093 · 2 repos pinned
3 igvfagent bench plan --seed → 16 analyses, one per figure panel
4 igvfagent paper-code pipeline → authors' scripts = reference
5 igvfagent port register → liu2025-open4gene, liu2025-gwas-loci
6 igvfagent bench verify-port → port vs R pscl::hurdle: P identical
7 igvfagent bench run → recompute Figs. 1–6 from tables
8 igvfagent figures redraw → Fig. 3D redrawn, side by side
9 igvfagent bench score → reproduction: incomplete, 11/16
loop 4–9 until every analysis is reproduced or blocked
Chain of tools
Output: Benchmarks/liu2025_kidney_multiome/README.md
Coverage 11/16 analyses reproduced; 43/46 checks pass
Ports liu2025-open4gene, liu2025-gwas-loci, each verified against the authors' own code
Exact ASE genes 8,573 / 8,725 / 10,398; bASA peaks 19,083; Open4Gene links 142,452; coding 1,363
Differs median peaks per gene 4 vs 8 (author re-checking); non-MHC variants +2.9%
Blocked RASQUAL, snASA, CARMA: individual-level genotypes, controlled access
Asks a table of exact data needed from the authors, ordered by how much each item unlocks
Output
c
User: Replicate this paper with IGVF Agent
Liu et al. 2025, Science
Fig. 3D: chromatin effect vs expression effect (R = 0.520, paper 0.51)
Effect size of Tubule ASE
Effect size of bASA
Gschwind et al. 2026, Nature
Fig. 2b,c: CRISPR benchmark AUPRC (ours from Supp. Table 3 vs paper)
paper
IGVF Agent
AUPRC (CRISPR benchmark)
0.66
0.56
0.74
0.61
ENCODE-rE2G
ABC
rE2G Extended
ABC (Ext. set)
Pan et al. 2026, Nat Methods
Fig. 1c: eSIG-Net finds disease mutations (ROC-AUC 0.908, paper 0.91)
True positive rate
False positive rate
eSIG-Net
DeepFE
D-SCRIPT
PPR
PLM-interact
SDNN
19 of 20 numbers within 5% of the paper
GWAS variants, P<5e-8
non-MHC variants
independent loci
ASE genes, tubule
ASE genes, glomerulus
ASE genes, union
bASA peaks
bASA SNPs
bASA vs ASE R
snASA SNPs
snASA in open chromatin (%)
PT snASA vs bASA rho
Open4Gene links
Open4Gene peaks
median peaks/gene
GWAS variants in peaks
coding variants
coding genes
convergent genes
validated (%)
-50%
IGVF Agent vs paper (%)
11 of 11 numbers within 5% of the paper
biosamples
cell types / tissues
elements per biosample
links per biosample
predicted links, total
genes per enhancer
enhancers per gene
AUPRC, rE2G
AUPRC, ABC
AUPRC, rE2G Extended
AUPRC, ABC (Extended set)
IGVF Agent vs paper (%)
12 of 12 numbers within 5% of the paper
accuracy, disease
ROC-AUC, disease
AP, disease
accuracy, population
ROC-AUC, population
AP, population
ROC-AUC vs structure
AP vs structure
FoldDock mutant acc.
FoldDock consistency (%)
FoldDock 17.6% (%)
FoldDock WT acc.
IGVF Agent vs paper (%)
11 of 16 analyses reproduced
11
3
2
5 of 5 analyses reproduced
5
11 of 16 analyses reproduced
11
1
4
reproduced (paper colour)
partial / not reproduced
blocked or pending